

# Efficient Inference for Inverse Reinforcement Learning and Dynamic Discrete Choice Models

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## Abstract

Inverse reinforcement learning (IRL) and dynamic discrete choice (DDC) models explain sequential decision-making through latent reward functions. Existing flexible IRL methods allow rich reward representations but typically do not support valid inference, whereas classical DDC approaches provide inference only under restrictive parametric structure. We develop a semiparametric framework for *debiased inverse reinforcement learning* in maximum-entropy IRL and Gumbel-shock DDC models. Our key identifying observation is that the log-behaviour policy can be treated as a pseudo-reward. It point-identifies policy value differences and, under a normalization constraint, the reward itself. This lets us express reward-dependent estimands as smooth functionals of the behaviour policy and transition kernel. We establish path-wise differentiability, derive efficient influence functions, and construct automatic debiased machine-learning estimators that permit flexible nuisance estimation while attaining  $\sqrt{n}$ -consistency, asymptotic normality, and semiparametric efficiency. The result is a computationally tractable framework connecting modern IRL with classical DDC inference.

**Keywords**— inverse reinforcement learning, dynamic discrete choice, semiparametric inference, debiased machine learning, policy evaluation

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# 1 Introduction

Behavioural data arise in robotics, human–computer interaction, healthcare, and economics. Inverse reinforcement learning (IRL) seeks to recover the latent reward or utility function underlying observed sequential decisions. It has been used to model expert driving and robotic control (e.g. [Abbeel and Ng, 2004](#); [Wulfmeier et al., 2016](#)), user behaviour in interactive systems (e.g. [Chandramohan et al., 2011](#); [Hossain et al., 2023](#)), and clinical decision-making in critical care (e.g. [Yu et al., 2019](#); [Bovenzi et al., 2025](#)). Because rewards summarize preferences or tradeoffs, they provide a basis for counterfactual analysis—for example, predicting how behaviour would change under new policies, constraints, or environments—that is not available from policy prediction alone.

A central difficulty is that IRL is inherently ill posed: many rewards can rationalize the same behavior ([Ng et al., 1999](#); [Ng and Russell, 2000](#); [Cao et al., 2021](#); [Skalse et al., 2023](#); [Skalse and Abate, 2024](#)). To make the problem tractable, IRL and dynamic discrete choice (DDC) models impose an optimality model linking rewards to behavior. Exact optimality implies deterministic policies and is often too rigid for observed human behavior. A foundational stochastic alternative in DDC adds Type-I extreme-value utility shocks, yielding choice probabilities of softmax form and a corresponding soft Bellman equation ([Rust, 1987](#); [Hotz and Miller, 1993](#); [Aguirregabiria and Mira, 2010](#)). The same softmax structure arises in modern maximum-entropy IRL by adding an entropy term to the agent’s objective, so actions with higher long-run utility are chosen with higher probability rather than deterministically ([Ziebart et al., 2008](#); [Ziebart, 2010](#); [Haarnoja et al., 2017](#)). Even under this structure, rewards remain identified only up to an equivalence class. Point identification therefore requires a normalization ([Rust, 1987](#); [Geng et al., 2020](#)), such as setting the reward of a reference action to zero.

Uncertainty quantification for reward-dependent summaries is central to IRL applications. These summaries include normalized reward averages, structural coefficients, and values of counterfactual decision rules (Kalouptside et al., 2021; Zielnicki et al., 2025). The problem is challenging because rewards are latent, identified only after normalization, and estimated through flexible nuisance components such as choice probabilities and transition laws. Existing methods largely split in two: classical DDC methods provide inference under restrictive parametric utility models (e.g. Rust, 1987; Hotz and Miller, 1993; Aguirregabiria and Mira, 2010), whereas flexible machine-learning approaches to IRL generally focus on reward estimation without valid uncertainty quantification for downstream functionals (e.g. Abbeel and Ng, 2004; Ramachandran and Amir, 2007; Ziebart et al., 2008; Levine et al., 2011; Ho and Ermon, 2016; Fu et al., 2018; Wulfmeier et al., 2016; Geng et al., 2020). We close this gap by developing a semiparametric framework for *debiased inverse reinforcement learning*.

## 1.1 Our contributions

We study reward identification and inference in maximum-entropy IRL and Gumbel-shock dynamic discrete choice models. Our main contributions are as follows:

1. We show that, in softmax IRL, the reward is recoverable from the log-behavior policy under an appropriate normalization. More generally, a broad class of reward-dependent estimands can be expressed as smooth functionals of the log-behavior policy and transition dynamics, reducing reward-based inference to inference on observable choice probabilities and the transition kernel.
2. We establish pathwise differentiability and derive efficient influence functions, thereby characterizing the semiparametric efficiency bounds for these targets. Under general affine reward normalizations, we show that the efficient influence function can be

obtained directly from the unnormalized case, yielding a general template for inference under normalization.

3. We construct automatic debiased machine-learning estimators (Chernozhukov et al., 2022; van der Laan et al., 2025a) that allow flexible nuisance estimation while achieving  $\sqrt{n}$ -consistency, asymptotic normality, and semiparametric efficiency.
4. We specialize the general theory to derive efficiency bounds and efficient estimators for several key IRL/DDC targets, including fixed-policy values, counterfactual softmax policy values, normalized-reward policy values, and structural coefficients of a normalized reward.

The paper proceeds as follows. Section 3 shows how policy-value differences, and under normalization the reward itself, can be identified from observed behavior. Sections 4.3 and 5 develop the semiparametric theory and automatic debiased machine-learning estimators for a broad class of smooth target functionals. Section 6 specializes these results to the main policy-value targets.

## 1.2 Related work

**Dynamic discrete choice models.** Classical DDC work studies identification and efficient estimation of finite-dimensional utility parameters using nested fixed-point, conditional choice probability, and simulation-based methods (Rust, 1987; Hotz and Miller, 1993; Hotz et al., 1994; Aguirregabiria and Mira, 2010). In contrast, we fix the softmax/Gumbel-shock structure of Rust (1987) and target reward-dependent functionals under nonparametric reward representations. To our knowledge, debiased estimation, efficient influence functions, and efficiency bounds for such functionals have not been developed in this literature. For identification, the closest connection is the conditional choice probability approach of Hotz and Miller (1993): under Gumbel shocks, log-odds of observed choices

identify differences in choice-specific value functions. Rather than using log-odds for value-difference identification, we use the log behavior policy together with the transition law to identify normalized rewards and reward-dependent functionals. An expanded related work is provided in Appendix D.

**Inverse reinforcement learning.** Classical IRL recovers parametric or linear rewards under hard optimality assumptions (e.g. [Ng and Russell, 2000](#); [Abbeel and Ng, 2004](#)), while more recent work uses maximum-entropy or adversarial formulations with flexible function approximators (e.g. [Ziebart et al., 2008, 2010](#); [Levine et al., 2011](#); [Ho and Ermon, 2016](#); [Fu et al., 2018](#); [Geng et al., 2020](#)). Several papers formalize the resulting reward ambiguity, including [Fu et al. \(2018\)](#), Theorem 1 of [Cao et al. \(2021\)](#), and Lemma 2 of [Geng et al. \(2020\)](#). Deep PQR ([Geng et al., 2020](#)) identifies normalized rewards under anchor-action constraints using a construction tailored to that setting: the reward is represented through the log behavior policy, an anchor-specific Bellman fixed point, and a second-stage conditional-mean regression. In contrast, our identification results express normalized rewards in terms of standard primitive objects—the log behavior policy, transition law, and policy-specific  $Q$ -functions—and apply to a broader class of affine normalization constraints, with anchor-action constraints as a special case. These works focus on reward recovery; our goal is inference for reward-dependent functionals.

**Off-policy evaluation and double reinforcement learning.** Debiased and doubly robust methods for off-policy evaluation estimate policy values when the reward is known (e.g. [Jiang and Li, 2016](#); [Thomas and Brunskill, 2016](#); [Tang et al., 2019](#); [Liu et al., 2018](#); [van Der Laan et al., 2018](#); [Kallus and Uehara, 2020](#); [Uehara et al., 2020](#); [Shi et al., 2021](#); [Kallus and Uehara, 2022](#); [Mehrabi and Wager, 2024](#)). Double Reinforcement Learning extends this perspective to more general functionals of the  $Q$ -function ([Kallus and Uehara, 2020, 2022](#); [van der Laan et al., 2025b](#)). Our framework is complementary: it treats the

reward as unknown, recovers it through the softmax behaviour model, and then performs semiparametric inference for the induced reward-dependent targets.

## 2 Preliminaries

### 2.1 Problem setup

We consider an infinite-horizon, time-homogeneous Markov decision process (MDP) with state space  $\mathcal{S}$ , action space  $\mathcal{A}$ , and reference measure  $\mu$  on  $\mathcal{S}$ . The initial state  $S_0 \in \mathcal{S}$  is drawn from density  $\rho_0$  with respect to  $\mu$ . At each time  $t \geq 0$ , the agent occupies state  $S_t$ , selects action  $A_t \in \mathcal{A}$  according to the behaviour policy  $\pi_0(\cdot \mid S_t)$ , receives latent reward  $r_0^\dagger(A_t, S_t)$ , and transitions to a next state  $S_{t+1} \in \mathcal{S}$  with conditional density  $k_0(\cdot \mid A_t, S_t)$  with respect to  $\mu$ . In the IRL setting, the analyst observes states and actions, but not rewards. We write  $P_0$  for the law of the one-step observation  $(S_0, A_0, S_1)$ , and  $\mathbb{P}_0$  for the induced law of the full trajectory.

The inferential task is to learn reward-dependent summaries, such as policy values, from observed state–action transitions. We observe  $\mathcal{D}_n := \{(S_i, A_i, S'_i)\}_{i=1}^n$ , a collection of  $n$  i.i.d. samples from  $P_0$ . Our analysis also accommodates temporal dependence through Markov-chain arguments under mixing conditions (Bibaut et al., 2021; Mehrabi and Wager, 2024). A fundamental challenge is that  $r_0^\dagger$  is not point identified from state–action data alone: many distinct rewards induce the same observable behaviour (Ng et al., 1999; Ng and Russell, 2000). IRL methods address this by imposing an optimality model linking the behaviour policy to the reward.

### 2.2 Soft optimality and the soft Bellman equation

We adopt the notion of *soft optimality*, which arises in Gumbel–shock-to-reward formulations (Rust, 1987; Hotz and Miller, 1993; Aguirregabiria and Mira, 2010) and in maximum-

entropy IRL (Ziebart et al., 2008, 2010; Haarnoja et al., 2017). Soft optimality replaces deterministic reward maximization with an entropy-regularized objective, yielding stochastic softmax decision rules. We first recall the classical (hard) value and  $Q$ -functions. For any policy  $\pi : \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$  and transition kernel  $k : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ , let  $\mathbb{E}_{k,\pi}$  denote expectation with respect to trajectories generated by the MDP with transition kernel  $k$ , policy  $\pi$ , and initial state density  $\rho_0$ .

For any function  $f : \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ , define the forward operators

$$\mathcal{P}_k f(a, s) := \int f(a, s') k(s' | a, s) \mu(ds'), \quad \pi f(s) := \sum_{a \in \mathcal{A}} \pi(a | s) f(a, s).$$

For a discount factor  $\gamma \in [0, 1)$  and reward function  $r : \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ , the (*hard*)  $Q$ -function is

$$q_{r,k}^{\pi,\gamma}(a, s) = \mathbb{E}_{k,\pi} \left[ \sum_{t=0}^{\infty} \gamma^t r(A_t, S_t) \mid A_0 = a, S_0 = s \right],$$

the unique bounded solution to Bellman's fixed-point equation

$$q_{r,k}^{\pi,\gamma}(a, s) = r(a, s) + \gamma \mathcal{P}_k V_{r,k}^{\pi,\gamma}(a, s), \quad V_{r,k}^{\pi,\gamma}(s) = \pi q_{r,k}^{\pi,\gamma}(s), \quad (1)$$

where  $V_{r,k}^{\pi,\gamma}$  is the associated (*hard*) *value function* (Bellman, 1954). Equivalently, it solves the inverse problem

$$\mathcal{T}_{k,\pi,\gamma}(q_{r,k}^{\pi,\gamma}) = r, \quad \mathcal{T}_{k,\pi,\gamma}(f) := f - \gamma \mathcal{P}_k \pi f,$$

where  $\mathcal{T}_{k,\pi,\gamma}$  is the Bellman operator. Since  $\gamma < 1$ , the operator  $\mathcal{T}_{k_0,\pi,\gamma}$  is invertible on the Banach space of state-action functions equipped with the supremum norm (Bellman, 1954), and we denote its inverse by  $\mathcal{T}_{k,\pi,\gamma}^{-1}$ . Throughout, when the discount factor is  $\gamma$ , we

omit explicit dependence on  $\gamma$  in our notation.

Entropy regularization augments the reward  $r_0(A_t, S_t)$  with an entropy bonus  $\tau \mathcal{H}(\pi(\cdot | S_t))$ , where  $\mathcal{H}(\pi(\cdot | s)) := -\sum_a \pi(a | s) \log \pi(a | s)$  is the Shannon entropy of the policy and  $\tau \in (0, \infty)$  is a regularization parameter. The value and  $Q$ -functions associated with this augmented reward are referred to as the *soft* value and  $Q$ -functions (Haarnoja et al., 2017). Since rewards are identified only up to scale, we set  $\tau = 1$  without loss of generality, interpreting rewards on a relative scale.

An agent is *soft optimal* (with discount factor  $\gamma$  and  $\tau = 1$ ) if its policy  $\pi_0$  maximizes the *soft value*, that is, the entropy-regularized expected return,

$$\pi \mapsto \mathbb{E}_{k_0, \pi} \left[ \sum_{t=0}^{\infty} \gamma^t \{ r_0^\dagger(A_t, S_t) + \mathcal{H}(\pi(\cdot | S_t)) \} \right],$$

over all policies  $\pi$  (Haarnoja et al., 2017). The resulting soft-optimal policy is the softmax (multinomial logit) distribution induced by the soft  $Q$ -function  $r_0^\dagger(a, s) + \gamma v_0^\dagger(a, s)$ :

$$\pi_0(a | s) = \frac{\exp\{r_0^\dagger(a, s) + \gamma v_0^\dagger(a, s)\}}{\sum_{a' \in \mathcal{A}} \exp\{r_0^\dagger(a', s) + \gamma v_0^\dagger(a', s)\}}, \quad (2)$$

where  $v_0^\dagger$  satisfies the soft Bellman fixed-point system

$$v_0^\dagger(a, s) = \int V_0^\dagger(s') k_0(s' | a, s) \mu(ds'), \quad V_0^\dagger(s) = \log \left( \sum_{a' \in \mathcal{A}} \exp\{r_0^\dagger(a', s) + \gamma v_0^\dagger(a', s)\} \right),$$

with  $V_0^\dagger$  the soft value function of  $\pi_0$  and  $v_0^\dagger$  its associated continuation value. Equivalently, in the maximum-entropy IRL formulation (Ziebart et al., 2008, 2010), the agent's policy maximizes the entropy of the trajectory distribution subject to matching the observed behaviour.<sup>1</sup> We take the discount factor  $\gamma$  as known. Identification of  $\gamma$  from observed

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<sup>1</sup>This yields a Boltzmann trajectory distribution,  $\mathbb{P}_0^{\pi_0}((S_0, A_0, \dots) = (s_0, a_0, \dots)) \propto$



choices—requiring additional variation beyond that in our setting—is analysed in [Abbring and Daljord \(2020\)](#).

Soft optimality also arises in dynamic discrete-choice (DDC) models ([Rust, 1987](#); [Hotz and Miller, 1993](#); [Aguirregabiria and Mira, 2010](#)), where the agent selects actions by maximizing a random utility  $A_t \in \arg \max_{a \in \mathcal{A}} \left\{ r_0^\dagger(a, S_t) + \gamma v_0^\dagger(a, S_t) + \varepsilon_t(a) \right\}$ , where, in the DDC model,  $v_0^\dagger(a, s)$  coincides with the classical choice-specific continuation value and  $\varepsilon_t(a)$  are i.i.d. Gumbel (Type I extreme value) shocks ([Rust, 1987](#); [Pitombeira-Neto et al., 2024](#)).

## 2.3 Notation

We introduce the notation and regularity conditions used throughout. Let  $\lambda := \# \otimes \mu$  denote the product measure on  $\mathcal{A} \times \mathcal{S}$ , where  $\#$  is the uniform measure on the finite action set  $\mathcal{A}$  and  $\mu$  is a finite dominating measure on  $\mathcal{S}$ . Thus, for any measurable  $B \subseteq \mathcal{A} \times \mathcal{S}$ ,  $\lambda(B) = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \int 1_B(a, s) \mu(ds)$ . Let  $\mathcal{H} := L^\infty(\lambda)$  denote the space of essentially bounded reward functions, and let  $\mathcal{K} \subset L^\infty(\mu \otimes \lambda)$  denote the space of essentially bounded transition kernels  $k$  (conditional densities of  $S' \mid (A, S)$  with respect to  $\mu$ ) satisfying  $c < k(s' \mid a, s) < C$  for  $\lambda$ -a.e.  $(a, s)$ . We equip  $\mathcal{H}$  and  $\mathcal{K}$  with the  $L^2$  norms

$$\|h\|_{\mathcal{H}} := \left( \int \sum_{a \in \mathcal{A}} h(a, s)^2 \mu(ds) \right)^{1/2}, \quad \|k\|_{\mathcal{K}} := \left( \int \sum_{a \in \mathcal{A}} k(s' \mid a, s)^2 \mu(ds') \lambda(a, ds) \right)^{1/2}.$$

For readability, we keep only the most frequently recurring conventions here. We write  $r_0^\dagger$  for the latent reward and  $r_0(a, s) := \log \pi_0(a \mid s)$  for the pseudo-reward. We use  $q$  and  $V$  for policy-specific  $Q$ - and state-value functions; when a scalar policy value is needed, we write  $V_{r,k}(\pi; \gamma)$ .

We make the simplifying assumption that the initial state distribution, behaviour policy, and transition dynamics are uniformly bounded away from zero and infinity.

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$$\exp\left(\sum_{t=0}^{\infty} \gamma^t r_0^\dagger(a_t, s_t)\right).$$

**(A1)** (*Bounded positivity*). There exist constants  $c, C \in (0, \infty)$  and  $\eta \in (0, 1)$  such that, for

$$\mu \otimes \lambda\text{-a.e. } (s, a, s') \in \mathcal{S} \times \mathcal{A} \times \mathcal{S}, c < \rho_0(s) < C, \quad \eta < \pi_0(a \mid s), c < k_0(s' \mid a, s) < C.$$

All results are stated under Assumption A1. Let  $\mathcal{M}$  denote the collection of distributions  $P(ds, da, ds') = \rho_P(s)\pi_P(a \mid s)k_P(s' \mid a, s)\mu(ds')\lambda(da, ds)$  satisfying this assumption. The boundedness requirement is imposed for technical convenience and can in principle be relaxed using suitable moment conditions on  $\rho_0$ ,  $\pi_0$ , and  $k_0$ .

### 3 Identification

We first characterize reward indeterminacy under soft optimality, then show that policy value differences remain identified, and finally impose a normalization to identify the reward itself.

#### 3.1 Partial identification of the reward

Without additional constraints, the reward  $r_0^\dagger$  is only partially identified: many rewards induce the same softmax policy for a soft-optimal agent (Cao et al., 2021). Its scale is also unidentified; by convention, we fix the entropy-regularization parameter at  $\tau = 1$ , which is equivalent to fixing the softmax temperature (Haarnoja et al., 2017). This subsection formalizes the resulting partial identification.

Formally, the partial identification of  $r_0^\dagger$  can be characterized through the softmax likelihood (Rust, 1987; Ziebart et al., 2008). The observed policy  $\pi_0$  maximizes this soft Bellman-constrained likelihood, so any reward  $r$  that induces the same softmax policy is also a solution of the population M-estimation problem

$$r \in \operatorname{argmin}_{r' \in \mathcal{H}} E_0 \left[ \log \left( \sum_{a \in \mathcal{A}} \exp \{ r'(a, S) + \gamma v_{r', k_0}(a, S) \} \right) - \{ r'(A, S) + \gamma v_{r', k_0}(A, S) \} \right], \quad (3)$$

where  $v_{r',k_0}$  satisfies the soft Bellman equation

$$v_{r',k_0}(a, s) = \int \log \left( \sum_{a' \in \mathcal{A}} \exp \{ r'(a', s') + \gamma v_{r',k_0}(a', s') \} \right) k_0(s' | a, s) \mu(ds'). \quad (4)$$

Any reward function that induces the same behaviour as  $r_0^\dagger$  has the potential-based reward-shaping form (Ng et al., 1999; Fu et al., 2018; Geng et al., 2020)

$$r(a, s) = r_0^\dagger(a, s) + c(s) - \gamma \int c(s') k_0(s' | a, s) \mu(ds'), \quad (5)$$

for a unique state-dependent potential  $c : \mathcal{S} \rightarrow \mathbb{R}$ . Under this transformation, the soft  $Q$ -functions satisfy  $q_{r,k_0}^{\text{soft}}(a, s) = q_{r_0^\dagger, k_0}^{\text{soft}}(a, s) + c(s)$ .

We begin with a trivial but informative solution to (3): the log-behaviour policy  $r_0(a, s) := \log \pi_0(a | s)$ . The soft Bellman constraint (4) is satisfied with  $v_{r_0, k_0} \equiv 0$ , because the normalization  $\sum_{a' \in \mathcal{A}} \pi_0(a' | s) = 1$  implies that the log-partition term vanishes,  $\log \sum_{a' \in \mathcal{A}} \exp \{ r_0(a', s) \} = 0$ .

**Lemma 1** (Trivial reward solution). *The log-behaviour policy  $r_0 = \log \pi_0$  is a solution to (3) and satisfies (5) with  $c(s) = -V_0^\dagger(s)$ .*

## 3.2 Identification of policy values

Although the reward is only partially identified, many quantities of practical interest remain point identified. Lemma 1 shows that behaviour determines  $r_0^\dagger$  only up to potential-based shaping. Such shifts change all policy values by the same constant and thus do not affect policy comparisons. Consequently, policy value differences remain identified, and the log-behaviour-policy reward  $r_0$  suffices for policy evaluation. The next result formalizes this fact.

**Theorem 2** (Behaviour policy identifies value differences). *Suppose that  $r_0^\dagger$  solves (3) for a discount factor  $\gamma \in [0, 1)$ . Then, for any policy  $\pi$ ,*

$$q_{r_0, k_0}^{\pi, \gamma}(a, s) = q_{r_0^\dagger, k_0}^{\pi, \gamma}(a, s) - V_0^\dagger(s), \quad V_{r_0, k_0}(\pi; \gamma) = V_{r_0^\dagger, k_0}(\pi; \gamma) - E_0[V_0^\dagger(S)].$$

*Hence, for any two policies  $\pi_1$  and  $\pi_2$ ,  $V_{r_0, k_0}(\pi_1; \gamma) - V_{r_0, k_0}(\pi_2; \gamma) = V_{r_0^\dagger, k_0}(\pi_1; \gamma) - V_{r_0^\dagger, k_0}(\pi_2; \gamma)$ .*

A subtle but important point is that the identification of policy value differences holds only for the discount factor  $\gamma$  that appears in the soft Bellman equation governing the agent's behaviour. To evaluate policies under a different discount factor  $\gamma'$ , the reward must typically be point identified. Section 3.3 shows how this can be achieved via a normalization constraint. Before turning to that result, we illustrate Theorem 2 in a counterfactual softmax setting.

**Example 1** (Identification for counterfactual softmax policies). *Let  $\mathcal{A}^* \subseteq \mathcal{A}$  and  $\tau^* \in (0, \infty)$ . For  $(r, k) \in \mathcal{H} \times \mathcal{K}$ , define the counterfactual softmax policy*

$$\pi_{r, k}^*(a \mid s) \propto \mathbb{1}\{a \in \mathcal{A}^*\} \exp\{(r(a, s) + \gamma v_{r, k}^*(a, s))/\tau^*\}, \quad (6)$$

*where  $v_{r, k}^*$  solves the counterfactual soft Bellman equation*

$$\begin{aligned} v_{r, k}^*(a, s) &= \int \Phi_{r, k}^*(s') k(s' \mid a, s) \mu(ds'), \\ \Phi_{r, k}^*(s) &:= \tau^* \log \left( \sum_{a' \in \mathcal{A}^*} \exp\left(\frac{1}{\tau^*}(r(a', s) + \gamma v_{r, k}^*(a', s))\right) \right). \end{aligned} \quad (7)$$

*Because  $\pi_{r, k}^*$  is invariant to potential-based reward shaping of the form (5), and Lemma 1 shows that  $r_0$  and  $r_0^\dagger$  differ by such a transformation, we have  $\pi_{r_0^\dagger, k_0}^* = \pi_{r_0, k_0}^*$ . Applying Theorem 2 to the pair of policies  $\pi_1 = \pi_{r_0, k_0}^*$  and  $\pi_2 = \pi_0$ , the value difference*

$V_{r_0^\dagger, k_0}(\pi_{r_0^\dagger, k_0}^\star; \gamma) - V_{r_0^\dagger, k_0}(\pi_0; \gamma)$  is identified using the log-behaviour-policy reward  $r_0$  in place of  $r_0^\dagger$ .  $\square$

### 3.3 Exact reward identification under normalization

The next theorem shows that, after normalization, the latent reward  $r_0^\dagger$  is exactly recoverable from the log-behaviour policy reward  $r_0$ . We fix a reference policy  $\nu$  and a known state-only anchor  $g : \mathcal{S} \rightarrow \mathbb{R}$ , and impose that the average reward in state  $s$  under  $\nu$  equals  $g(s)$ :

$$\nu r_0^\dagger(s) := \sum_{\tilde{a} \in \mathcal{A}} \nu(\tilde{a} \mid s) r_0^\dagger(\tilde{a}, s) = g(s), \quad P_0\text{-a.e. } s. \quad (8)$$

This constraint anchors the free state potential in (5) and covers common normalizations, including anchor-action, statewise sum-to-zero, and value-based constraints (Hotz and Miller, 1993; Bajari et al., 2010; Kallus and Udell, 2016; Geng et al., 2020). For example, requiring  $V_{r_0^\dagger, k}^{\nu, \gamma}(s) = h(s)$  for all  $s$  is equivalent, by the Bellman equation, to choosing  $g(s) = h(s) - \gamma \sum_{a \in \mathcal{A}} \nu(a \mid s) \int h(\tilde{s}) k(d\tilde{s} \mid a, s)$ . The pair  $(\nu, g)$  can also be chosen using auxiliary information, such as outcomes that determine a reference reward level (Kallus et al., 2018). For instance, with auxiliary state-reward data from the agent, one could take  $\nu = \pi$  and  $g(s) = E_0\{r(A, S) \mid S = s\}$ .

The following identification result shows that the normalized reward can be expressed through the  $Q$ -function  $q_{r_0 - g, k_0}^{\nu, \gamma} = \mathcal{T}_{k_0, \nu, \gamma}^{-1}(r_0 - g)$  of the normalization policy  $\nu$ .

**Theorem 3** (Normalization identifies the reward itself). *Let  $\nu$  be a policy. There exists a unique solution  $r_{0, \nu, \gamma, g} \in \mathcal{H}$  to (3) that satisfies  $\nu r_{0, \nu, \gamma, g}(s) = g(s)$  for  $P_0$ -a.e.  $s \in \mathcal{S}$ .*

Moreover,

$$r_{0,\nu,\gamma,g}(a, s) = g(s) + q_{r_0-g,k_0}^{\nu,\gamma}(a, s) - \sum_{\tilde{a} \in \mathcal{A}} \nu(\tilde{a} \mid s) q_{r_0-g,k_0}^{\nu,\gamma}(\tilde{a}, s),$$

$$v_{0,\nu,\gamma,g}(a, s) = \frac{1}{\gamma} (r_0(a, s) - g(s) - q_{r_0-g,k_0}^{\nu,\gamma}(a, s)),$$

where  $v_{0,\nu,\gamma,g}$  solves the soft Bellman equation with reward  $r_{0,\nu,\gamma,g}$ .

Thus, if  $r_0^\dagger$  satisfies (8), it is identified by  $r_{0,\nu,\gamma,g}$ . Inference under normalization therefore reduces to estimating the observable pseudo-reward  $r_0 = \log \pi_0$  and the Bellman quantity  $q_{r_0-g,k_0}^{\nu,\gamma}$ , which motivates the next section.

## 4 Debiased Inverse Reinforcement Learning

We now move from identification to inference. We first define a generic target functional built from the pseudo-reward and transition kernel, then derive its efficient influence function and bias expansion, and finally specialize these objects to the motivating examples below.

### 4.1 Statistical objective

The identification results in Section 3 imply that the log-behaviour policy  $r_0 = \log \pi_0$  can be used as a *pseudo-reward* for many IRL targets. Policy value differences are identified directly from  $r_0$  by Theorem 2, and Theorem 3 shows that normalized reward functionals can also be recovered from  $(r_0, k_0)$ . We therefore study smooth real-valued functionals of the form

$$\psi_0 := E_0[m(A, S, r_0, k_0)],$$

where  $(r, k) \mapsto m(\cdot, r, k)$  is a smooth map from  $\mathcal{H} \times \mathcal{K}$  to  $L^2(\lambda)$ . Formally,  $\psi_0 = \Psi(P_0)$  corresponds to the parameter  $\Psi : \mathcal{M} \rightarrow \mathbb{R}$  defined for  $P \in \mathcal{M}$  by  $\Psi(P) := E_P[m(A, S, r_P, k_P)]$ ,

where  $k_P(s' \mid a, s) := \frac{dP_{S'|A,S}}{d\mu}(s' \mid a, s)$  is the state transition kernel, and  $r_P(a, s) = \log P(A = a \mid S = s)$ .

## 4.2 Motivating examples

Our framework accommodates a wide range of counterfactual targets in IRL. We record the main targets used in the sequel: three policy-value parameters and one structural normalized-reward parameter.

**Example 1a** (Policy value). *The policy value  $\psi_0 := E_0[V_{r_0, k_0}^{\pi, \gamma}(S)]$  of a policy  $\pi$  with discount factor  $\gamma$  corresponds to the map  $m(a, s, r, k) = V_{r, k}^{\pi, \gamma}(s)$ .  $\square$*

The same framework also covers interventions on the agent’s behaviour or environment.

**Example 1b** (Value of a soft-optimal agent). *Consider the counterfactual softmax policy  $\pi_{r_0, k_0}^*$  defined in (6), obtained by optimizing the pseudo-reward  $r_0$  under kernel  $k_0$  with action set  $\mathcal{A}^*$  and entropy parameter  $\tau^*$ . Unlike Example 1a, this policy depends on both  $r_0$  and  $k_0$ . The policy value  $\psi_0$  fits our framework through the map  $m(a, s, r, k) := V_{r, k}^*(s)$ , where  $V_{r, k}^*$  denotes the value function of  $\pi_{r, k}^*$ .  $\square$*

A closely related counterfactual keeps the pseudo-reward target fixed but replaces the transition kernel by a known  $k^*$ , as in simulated or redesigned environments (e.g. [Abbeel and Ng, 2004](#); [Ziebart et al., 2008](#); [Wulfmeier et al., 2016](#); [Fu et al., 2018](#)).

**Example 1c** (Policy values in counterfactual environments). *The value of the soft-optimal agent that maximizes the pseudo-reward  $r_0$  in an environment with known transition kernel  $k^*$  corresponds to  $m(a, s, r, k) := V_{r, k^*}^*(s)$ , where  $V_{r, k^*}^*$  denotes the value function of  $\pi_{r, k^*}^*$ .  $\square$*

Under a normalization constraint, functionals of the true reward  $r_0^\dagger$  also fall within our framework. In these settings, the target is identified by the  $\nu$ -normalized reward  $\bar{r}_0 := r_{0, \nu, \gamma, g}$ .

**Example 2** (Policy values under reward normalization). *Suppose we are interested in  $\psi_0 = E_0[\tilde{m}(A, S, \bar{r}_0, k_0)]$ , a feature of the  $\nu$ -normalized reward  $\bar{r}_0$ . By Theorem 3, this can be expressed as  $\psi_0 = E_0[m(A, S, r_0, k_0)]$ , where*

$$m(a, s, r, k) := \tilde{m}(a, s, r_{r,k,\nu,\gamma,g}, k), \quad r_{r,k,\nu,\gamma,g}(a, s) := g(s) + q_{r-g,k}^{\nu,\gamma}(a, s) - V_{r-g,k}^{\nu,\gamma}(s),$$

and  $r_{r,k,\nu,\gamma,g} = g + (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1}(r - g)$  denotes the  $\nu$ -normalized reward with anchor  $g$ . This includes fixed-policy values or softmax values built with a target discount  $\gamma'$  that may differ from the normalization discount  $\gamma$ .  $\square$

**Example 3** (Structural coefficients under reward normalization). *Let  $x : \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}^p$  be a known bounded feature map and write  $\bar{r}_0 := r_{0,\nu,\gamma,g}$ . If  $G_0 := E_0\{x(A, S)x(A, S)^\top\}$  is nonsingular, the coefficient in the linear projection of the normalized reward onto  $x$  is*

$$\theta_0 := G_0^{-1} E_0\{x(A, S)\bar{r}_0(A, S)\}.$$

For example, in Rust's bus replacement model, a coordinate of  $x(a, s)$  may indicate replacement, in which case the corresponding coordinate of  $\theta_0$  is a replacement-cost coefficient.  $\square$

### 4.3 Statistical efficiency and functional bias expansion

A key step in our methodology is to establish the pathwise differentiability of  $\Psi$  and derive its efficient influence function (EIF). In semiparametric statistics, the EIF determines the generalized Cramér–Rao lower bound for  $\Psi$  and forms the basis for constructing debiased, efficient estimators (Bickel et al., 1993; Laan and Robins, 2003; Van der Laan et al., 2011; Chernozhukov et al., 2018).

The next conditions ensure that  $E_0[m(A, S, r, k)]$  is first-order expandable in  $r$  and  $k$ . The *reward tangent space*  $T_{\mathcal{H}}$  is defined as the completion of  $\mathcal{H}$  under the inner product



$\langle h_1, h_2 \rangle_{\mathcal{H},0} := E_0[h_1(A, S)h_2(A, S)]$ . We define the *kernel tangent space at  $k$*  by

$$T_{\mathcal{K}}(k) := \left\{ \beta : \int \beta(s', a, s)^2 k(s' | a, s) \mu(ds') < \infty \text{ and } \int \beta(s', a, s) k(s' | a, s) \mu(ds') = 0 \text{ for } \lambda\text{-a.e. } (a, s) \right\},$$

and set  $T_{\mathcal{K}} := T_{\mathcal{K}}(k_0)$ , equipped with the inner product  $\langle \beta_1, \beta_2 \rangle_{\mathcal{K},0} := E_0[\beta_1(S', A, S)\beta_2(S', A, S)]$ .

Intuitively,  $T_{\mathcal{K}}$  consists of allowable infinitesimal kernel scores that preserve the normalization of conditional densities under  $k(\cdot | a, s)\mu(ds')$ . For the rate statements below, we also write  $\|h\|_{\mathcal{H},r_0} := \{E_0[h(A, S)^2]\}^{1/2}$ ; this is the same norm as  $\|h\|_{\mathcal{H},0}$ , with the subscript emphasizing that the action law is  $\pi_0 = \exp(r_0)$ .

**(C1)** (*Functional differentiability*) There exist Gâteaux derivatives  $\partial_r m(\cdot, r_0, k_0) : T_{\mathcal{H}} \rightarrow L^2(\lambda)$  and  $\partial_k m(\cdot, r_0, k_0) : T_{\mathcal{K}} \rightarrow L^2(\lambda)$  at  $(r_0, k_0)$  such that, for all  $\alpha \in T_{\mathcal{H}}$  and  $\beta \in T_{\mathcal{K}}$ ,

$$\left. \frac{d}{dt} E_0[m(A, S, r_0 + t\alpha, (1 + t\beta)k_0)] \right|_{t=0} = E_0[\partial_r m(A, S, r_0, k_0)(\alpha) + \partial_k m(A, S, r_0, k_0)(\beta)].$$

**(C2)** (*Continuity of derivatives*) The linear functionals  $E_0[\partial_r m(A, S, r_0, k_0)(\cdot)]$  and  $E_0[\partial_k m(A, S, r_0, k_0)(\cdot)]$  are continuous, in the sense that

$$\sup_{\alpha \in T_{\mathcal{H}}} \frac{E_0[\partial_r m(A, S, r_0, k_0)(\alpha)]}{\|\alpha\|_{\mathcal{H},0}} < \infty, \quad \sup_{\beta \in T_{\mathcal{K}}} \frac{E_0[\partial_k m(A, S, r_0, k_0)(\beta)]}{\|\beta\|_{\mathcal{K},0}} < \infty.$$

**(C3)** (*Lipschitz continuity*) The map  $(r, k) \mapsto E_0[m(A, S, r, k)]$  is locally Lipschitz continuous at  $(r_0, k_0)$  with respect to  $(\mathcal{H}, \|\cdot\|_{\mathcal{H},0}) \times (\mathcal{K}, \|\cdot\|_{\mathcal{K}})$ .

Conditions **C1-C3** hold if the map  $(r, k) \mapsto E_0[m(A, S, r, k)]$  is Fréchet differentiable at  $(r_0, k_0)$  with respect to the product norm  $\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}}$ ; see Lemma 6 in Appendix F. For additive Fréchet derivatives in the kernel argument, we write  $\tilde{T}_{\mathcal{K}}$  for the ambient normed linear space of additive kernel perturbations, equipped with  $\|\cdot\|_{\mathcal{K}}$ , and denote the corre-

sponding additive derivative by  $\tilde{\partial}_k$ . Thus  $\partial_k$  is reserved for score derivatives on  $T_{\mathcal{K}}$ , while  $\tilde{\partial}_k$  acts on additive directions  $g \in \tilde{T}_{\mathcal{K}}$ .

The EIF depends on two additional nuisance functions: the Riesz representers of the partial functional derivatives of  $(r, k) \mapsto E_0[m(A, S, r, k)]$ . Let  $\alpha_0 \in T_{\mathcal{H}}$  and  $\beta_0 \in T_{\mathcal{K}}$  denote these representers with respect to  $\langle \cdot, \cdot \rangle_{\mathcal{H},0}$  and  $\langle \cdot, \cdot \rangle_{\mathcal{K},0}$ , so that

$$E_0[\partial_r m(A, S, r_0, k_0)(\alpha)] = \langle \alpha_0, \alpha \rangle_{\mathcal{H},0}, \quad (9)$$

$$E_0[\partial_k m(A, S, r_0, k_0)(\beta)] = \langle \beta_0, \beta \rangle_{\mathcal{K},0}. \quad (10)$$

**Theorem 4** (Pathwise differentiability). *Under C1-C3, the parameter  $\Psi : \mathcal{M} \rightarrow \mathbb{R}$  is pathwise differentiable at  $P_0$ . Its efficient influence function is*

$$\chi_0(s, a, s') := \alpha_0(a, s) - \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) \alpha_0(\tilde{a}, s) + \beta_0(s, a, s') + m(a, s, r_0, k_0) - \Psi(P_0),$$

where  $\alpha_0$  and  $\beta_0$  are the Riesz representers defined in (9)–(10).

The EIF  $\chi_0$  determines the generalized Cramér–Rao lower bound for  $\Psi$ . Its kernel component follows from existing EIF results for offline reinforcement learning. Specifically, if the rewards  $\{r_0(A_i, S_i)\}_{i=1}^n$  were observed, then  $\beta_0(S_i, A_i, S'_i) + m(A_i, S_i, r_0, k_0) - \Psi(P_0)$  would be the nonparametric EIF for the known-reward target  $P \mapsto E_P\{m(A, S, r_0, k_P)\}$ . Thus, in each IRL application, it remains only to derive the reward-side Riesz representer  $\alpha_0$  for the map  $r \mapsto E_0\{\partial_r m(A, S, r, k_0)\}$ . For example, EIFs for policy values are given in van Der Laan et al. (2018); Kallus and Uehara (2020, 2022), and van der Laan et al. (2025b) derives EIFs for general continuous linear functionals of the  $Q$ -function. Although those works allow additive reward noise, the EIF for continuous linear functionals of the  $Q$ -function is unchanged because the reward-noise score is orthogonal to the target-relevant

tangent space ([Bickel et al., 1993](#); [Laan and Robins, 2003](#)).

The EIF characterizes the first-order behaviour of plug-in estimators of  $\psi_0$ . To formalize this, we introduce the score

$$\varphi_{r,k}(s', a, s; \alpha, \beta) := \alpha(a, s) - \sum_{\tilde{a} \in \mathcal{A}} \exp\{r(\tilde{a}, s)\} \alpha(\tilde{a}, s) + \beta(s', a, s),$$

which provides the linear approximation of  $E_0[m(A, S, r, k)]$  around  $(r_0, k_0)$ . To analyze the bias of this approximation, define the remainder

$$\text{Rem}_0(r_0, k_0; r, k) := E_0[m(A, S, r, k)] - E_0[m(A, S, r_0, k_0)] - E_0[\varphi_{r_0, k_0}(\cdot; \alpha_0, \beta_0)].$$

The next theorem establishes a von Mises expansion for this decomposition ([Mises, 1947](#)).

**Theorem 5** (von Mises expansion). *Assume [C1](#). Let  $r \in \mathcal{H}$ ,  $k \in \mathcal{K}$ ,  $\alpha \in T_{\mathcal{H}}$ , and  $\beta \in T_{\mathcal{K}}(k)$ , and suppose  $\|r\|_{L^\infty(\lambda)} + \|r_0\|_{L^\infty(\lambda)} < M$  for some  $M \in (0, \infty)$ . Then*

$$\begin{aligned} E_0[m(A, S, r, k)] - E_0[m(A, S, r_0, k_0)] &+ \int \varphi_{r,k}(s', a, s; \alpha, \beta) P_0(ds, da, ds') \\ &= -\langle r - r_0, \alpha - \alpha_0 \rangle_{\mathcal{H}, 0} + O(\|r - r_0\|_{\mathcal{H}}^2) \\ &+ O\left(\|\beta_0 - \beta\|_{L^2(P_0)} \left\| \frac{k - k_0}{k_0} \right\|_{L^2(P_0)}\right) + \text{Rem}_0(r_0, k_0; r, k), \end{aligned}$$

where the implicit constants depend only on  $M$ .

It is often possible to show that the linearization remainder is second order, for example  $\text{Rem}_0(r_0, k_0; r, k) \lesssim \|k - k_0\|_{\mathcal{K}}^2 + \|r - r_0\|_{\mathcal{H}}^2$ . For [Example 1a](#), this follows from [Lemma 10](#) in [Appendix H](#).

## 4.4 Functionals of $\nu$ -normalized reward

Suppose we have already derived the EIF of a functional of the pseudo-reward  $r_0$  using Theorem 4, and we now seek the EIF of the corresponding functional of the  $\nu$ -normalized reward  $r_{0,\nu,\gamma,g}$  from Theorem 3. The next result shows that the EIF can be obtained directly from the unnormalized case via a simple transformation of the associated Riesz representers.

Consider the functional  $\psi_0 = E_0[\tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)]$  from Example 2. Let  $\tilde{\alpha}_0$  and  $\tilde{\beta}_0$  be the Riesz representers of the partial derivatives of the map  $(r, k) \mapsto E_0[\tilde{m}(A, S, r, k)]$  at  $(r_{0,\nu,\gamma,g}, k_0)$ , defined by the property that for all  $\alpha \in T_{\mathcal{H}}$  and  $\beta \in T_{\mathcal{K}}$ ,

$$E_0[\partial_r \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)[\alpha]] = \langle \tilde{\alpha}_0, \alpha \rangle_{\mathcal{H},0},$$

$$E_0[\partial_k \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)[\beta]] = \langle \tilde{\beta}_0, \beta \rangle_{\mathcal{K},0}.$$

The Riesz representers are defined analogously to (9) and (10), the only difference being that the partial derivatives are now evaluated at  $(r_{0,\nu,\gamma,g}, k_0)$  rather than  $(r_0, k_0)$ . The following theorem assumes Conditions C1\*-C3\*, which adapt Conditions C1-C3 to the map  $\tilde{m}$  evaluated at  $(r_{0,\nu,\gamma,g}, k_0)$ . Since these conditions are direct analogues, they are stated in Appendix A.1.

**Theorem 6** (Pathwise differentiability for normalized rewards). *Suppose C1\*-C3\*, and define  $m(a, s, r, k) := \tilde{m}(a, s, r_{r,k,\nu,\gamma,g}, k)$ . Then the parameter  $\Psi : P \mapsto E_P[m(A, S, r_P, k_P)]$  is pathwise differentiable with EIF  $\chi_0$ , given in Theorem 4, where*

$$\alpha_0(a, s) := \tilde{\alpha}_0(a, s) - \frac{\nu(a \mid s)}{\pi_0(a \mid s)} \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) \tilde{\alpha}_0(\tilde{a}, s);$$

$$\beta_0(s' \mid a, s) := \tilde{\beta}_0(s' \mid a, s) + \alpha_0(a, s) \left\{ r_0(a, s) - g(s) + \gamma V_{r_0-g, k_0}^{\nu, \gamma}(s') - q_{r_0-g, k_0}^{\nu, \gamma}(a, s) \right\}.$$

Equivalently, Theorem 6 gives a conditional influence operator for the normalized reward itself. For any downstream functional whose derivative with respect to the normalized reward has Riesz representer  $w \in T_{\mathcal{H}}$ , define

$$\mathcal{A}_\nu w(a, s) := w(a, s) - \frac{\nu(a \mid s)}{\pi_0(a \mid s)} \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) w(\tilde{a}, s)$$

and

$$\Delta_\nu(s', a, s) := r_0(a, s) - g(s) + \gamma V_{r_0-g, k_0}^{\nu, \gamma}(s') - q_{r_0-g, k_0}^{\nu, \gamma}(a, s).$$

Then the contribution to the EIF coming only from estimating  $r_{0, \nu, \gamma, g}$  is

$$\mathcal{I}_\nu(w)(s', a, s) := \mathcal{A}_\nu w(a, s) - \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) \mathcal{A}_\nu w(\tilde{a}, s) + \mathcal{A}_\nu w(a, s) \Delta_\nu(s', a, s).$$

Thus, to compute the EIF for any smooth functional  $E_0\{\tilde{m}(A, S, r_{0, \nu, \gamma, g}, k_0)\}$ , one may first compute the ordinary Riesz representers  $(w, \tilde{\beta})$  of  $\tilde{m}$  as if the normalized reward were observed, and then use

$$\chi_0 = \mathcal{I}_\nu(w) + \tilde{\beta}_0(S', A, S) + \tilde{m}(A, S, r_{0, \nu, \gamma, g}, k_0) - \psi_0.$$

This separates the reusable normalized-reward influence calculation from the functional-specific derivative  $w$ .

The simple structure of the EIF follows from the following representation.

**Lemma 7** (Representation of reward averages under normalization). *For any  $w \in L^2(P_0)$ ,*

$$E_0[w(A, S) r_{0, \nu, \gamma, g}(A, S)] = E_0 \left[ w(A, S) \left\{ g(S) + r_0(A, S) - \sum_{a \in \mathcal{A}} \nu(a \mid S) r_0(a, S) \right\} \right].$$

Hence, weighted averages of  $r_{0, \nu, \gamma, g}$ , with possibly signed weights  $w$ , reduce to the reward-

free term  $E_0\{w(A, S)g(S)\}$  and an average of  $r_0$  after subtracting its  $\nu$ -mean at the same state. Consequently, linear summaries such as the best linear predictor of  $r_{0,\nu,\gamma,g}(A, S)$  can be identified from the pseudo-reward.

## 5 Automatic debiased estimation and inference

### 5.1 General form of the estimator

Let  $r_n$  and  $k_n$  denote estimators of the pseudo-reward  $r_0$  (the log-behaviour policy) and the transition kernel  $k_0$ , respectively. The pseudo-reward  $r_0$  can be learned from state-action data  $\{(S_i, A_i)\}_{i=1}^n$  via probabilistic classification, while  $k_0$  may be estimated using conditional density methods. A natural estimator of  $\psi_0$  is the plug-in estimator  $n^{-1} \sum_{i=1}^n m(A_i, S_i, r_n, k_n)$ . Under flexible nuisance estimation, this plug-in approach can exhibit substantial bias. We therefore use a one-step influence-function-based correction.

Let  $\alpha_n \in \mathcal{H}$  and  $\beta_n \in L^2(P_0)$  estimate the representers  $\alpha_0$  and  $\beta_0$  from Section 4.3. We take  $\beta_n$  to be centered with respect to  $k_n$ , so that  $\int \beta_n(s', a, s) k_n(s' \mid a, s) \mu(ds') = 0$ ; this can always be enforced post hoc. We propose the one-step estimator  $\psi_n := P_n\{m(\cdot, r_n, k_n) + \varphi_{r_n, k_n}(\cdot; \alpha_n, \beta_n)\}$ , or, equivalently,

$$\psi_n = \frac{1}{n} \sum_{i=1}^n m(A_i, S_i, r_n, k_n) + \frac{1}{n} \sum_{i=1}^n \left[ \alpha_n(A_i, S_i) - \sum_{a \in \mathcal{A}} \exp\{r_n(a, S_i)\} \alpha_n(a, S_i) + \beta_n(S'_i, A_i, S_i) \right]. \quad (11)$$

The second term applies the EIF from Theorem 4 to correct the first-order bias of the plug-in estimator (Bickel et al., 1993). For policy value parameters, the kernel representer  $\beta_0$  coincides with the usual transition-kernel component of the EIF in off-policy evaluation, and can be estimated using existing DRL methods. The reward representer can often be derived analytically or estimated directly by *Riesz regression* (Chernozhukov et al., 2022;

van der Laan et al., 2025a), using the population risk

$$\alpha \mapsto E_0 \left[ \alpha(A, S)^2 - 2 \partial_r m(A, S, r_0, k_0)(\alpha) \right].$$

Because (11) is agnostic to the choice of  $m$ , we refer to this procedure as an *automatic* DML estimator.

When  $\psi_0$  depends on value- or  $Q$ -functions induced by  $k_0$ , these higher-level nuisances can often be estimated directly, avoiding explicit estimation of  $k_0$ . The next section uses this reparameterization for the main policy-value examples.

## 5.2 Asymptotic theory

The next theorem establishes that the proposed estimator  $\psi_n$  is asymptotically linear with influence function  $\chi_0$  and is therefore efficient for  $\psi_0$ . We impose the following conditions.

**(C4) (*Sample splitting and boundedness*)** The estimators  $r_n$ ,  $k_n$ ,  $\alpha_n$ , and  $\beta_n$  are obtained from data independent of  $\mathcal{D}_n$ . Moreover,  $\|m(\cdot, r_n, k_n) + \varphi_{r_n, k_n}(\cdot; \alpha_n, \beta_n)\|_{L^2(P_0)} = O_P(1)$  and  $\|m(\cdot, r_0, k_0) + \varphi_{r_0, k_0}(\cdot; \alpha_0, \beta_0)\|_{L^2(P_0)} = O(1)$ .

**(C5) (*Nuisance consistency and estimation rates*)**

$$(a) \quad \|m(\cdot, r_n, k_n) + \varphi_{r_n, k_n}(\cdot; \alpha_n, \beta_n) - m(\cdot, r_0, k_0) - \varphi_{r_0, k_0}(\cdot; \alpha_0, \beta_0)\|_{L^2(P_0)} = o_P(1)$$

$$(b) \quad \|r_n - r_0\|_{\mathcal{H}} = o_P(n^{-1/4}) \text{ and } \|r_n - r_0\|_{\mathcal{H}, r_0} \|\alpha_n - \alpha_0\|_{\mathcal{H}, r_0} = o_P(n^{-1/2}).$$

$$(c) \quad \left\| \frac{k_n - k_0}{k_0} \right\|_{L^2(P_0)} \|\beta_n - \beta_0\|_{L^2(P_0)} = o_P(n^{-1/2}) \text{ and } \text{Rem}_0(r_0, k_0; r_n, k_n) = o_P(n^{-1/2}).$$

**Theorem 8** (Asymptotic linearity and efficiency). *Suppose Conditions C4–C5 hold. Then the autoDML estimator  $\psi_n$  defined in (11) satisfies*

$$\psi_n - \psi_0 = \frac{1}{n} \sum_{i=1}^n \chi_0(S_i, A_i, S'_i) + o_P(n^{-1/2}).$$

Consequently,  $\sqrt{n}(\psi_n - \psi_0) \rightsquigarrow N(0, \sigma_0^2)$ , where  $\sigma_0^2 = \text{Var}_{P_0}(\chi_0(S, A, S'))$ . Thus  $\psi_n$  is nonparametric efficient for  $\psi_0$ .

The conditions above are standard in debiased machine learning. Condition C4 enforces independence between nuisance estimation and evaluation; this can be relaxed to cross-fitting, where the sample is partitioned and nuisance functions are estimated on held-out folds (Van der Laan et al., 2011; Chernozhukov et al., 2018). Condition C4 guarantees uniform  $L^2$  boundedness of the true and estimated uncentered influence functions. Condition (a) requires consistency of this estimated uncentered influence function, typically ensured when the nuisance estimators are consistent. Condition C5 specifies the convergence rates needed for the von Mises remainder to be second order. In particular, part (c) yields a form of partial double robustness: slower convergence of  $\alpha_n$  can be offset by faster convergence of  $r_n$ . In our examples, the linearization remainder often satisfies  $\text{Rem}_0(r_0, k_0; r, k) = O(\|r - r_0\|_{\mathcal{H}}^2 + \|k - k_0\|_{\mathcal{K}}^2)$ .

## 6 Applications

This section specializes the generic theory to the three policy-value targets developed in the main text: the value of a fixed policy, the value of a fixed policy under reward normalization, and the value of a counterfactual soft-optimal policy. The corresponding normalized softmax and structural coefficient examples are collected in Appendix A.

### 6.1 Efficient influence functions for example parameters

We begin with Example 1a, which concerns the value of a fixed policy  $\pi$ . Our analysis relies on an overlap condition for the stationary state-action distribution, ensuring that the operator  $\mathcal{T}_{k_0, \pi}$  has a bounded inverse on  $L^2(\lambda)$ . For a given policy  $\pi$ , let  $p_k^{\pi, \gamma}$  denote a stationary state- $\mu$ -density for the MDP with policy  $\pi$  and kernel  $k$ . That is, for every



$f \in L^\infty(\mu)$ ,  $\int f(s') p_k^{\pi, \gamma}(s') \mu(ds') = \iint f(s') k'_\pi(s' \mid s) \mu(ds') p_k^{\pi, \gamma}(s) \mu(ds)$ , where  $k'_\pi(s' \mid s) := \sum_{a \in \mathcal{A}} \pi(a \mid s) k(s' \mid a, s)$ .

**(D1)** (*Stationary overlap for  $\pi$* ) There exist constants  $b, B \in (0, \infty)$  such that  $b \leq \frac{p_{k_0}^{\pi, \gamma}}{\rho_0}(s)$ .  
 $\frac{\pi(a \mid s)}{\pi_0(a \mid s)} \leq B$  for  $\lambda$ -a.e.  $(a, s)$ .

In what follows, denote the (unnormalized) discounted state–action occupancy density ratio by

$$d_0^{\pi, \gamma}(a, s) := \rho_0^{\pi, \gamma}(s) \frac{\pi(a \mid s)}{\pi_0(a \mid s)}, \quad \rho_0^{\pi, \gamma}(s) := \frac{1}{\rho_0(s)} \sum_{t=0}^{\infty} \gamma^t \frac{d\mathbb{P}_{k_0, \pi}}{d\mu}(S_t = s), \quad (12)$$

where  $\rho_0^{\pi, \gamma}$  is the associated state occupancy ratio. This state–action occupancy ratio characterizes the policy value through the identity  $E_0[V_{r_0, k_0}^{\pi, \gamma}(S)] = E_0[d_0^{\pi, \gamma}(A, S) r_0(A, S)]$ . In this sense, it plays the same role in offline policy evaluation as the inverse propensity score does in causal inference.

**Theorem 9** (EIF for known policy). *Let  $m(a, s, r, k) := V_{r, k}^{\pi, \gamma}(s)$ . Under [D1](#), the parameter  $\Psi : P \mapsto E_P[V_{r_P, k_P}^{\pi, \gamma}(S)]$  is pathwise differentiable at  $P_0$  with efficient influence function*

$$\begin{aligned} \chi_0(s, a, s') &= \rho_0^{\pi, \gamma}(s) \left( \frac{\pi(a \mid s)}{\pi_0(a \mid s)} - 1 \right) \\ &\quad + d_0^{\pi, \gamma}(a, s) \left\{ r_0(a, s) + \gamma V_{r_0, k_0}^{\pi, \gamma}(s') - q_{r_0, k_0}^{\pi, \gamma}(a, s) \right\} \\ &\quad + V_{r_0, k_0}^{\pi, \gamma}(s) - \Psi(P_0). \end{aligned}$$

In [Theorem 9](#), the representers from [Theorem 4](#) take the explicit forms  $\alpha_0(a, s) = \rho_0^{\pi, \gamma}(s) \frac{\pi(a \mid s)}{\pi_0(a \mid s)}$  and  $\beta_0(s, a, s') = d_0^{\pi, \gamma}(a, s) \{r_0(a, s) + \gamma V_{r_0, k_0}^{\pi, \gamma}(s') - q_{r_0, k_0}^{\pi, \gamma}(a, s)\}$ . Thus the EIF combines a density-ratio term from the pseudo-reward with the familiar temporal-difference term from off-policy evaluation ([van Der Laan et al., 2018](#); [Kallus and Uehara, 2020, 2022](#); [van der](#)

Laan et al., 2025b).

We next consider Example 1b, concerning the value of the counterfactual softmax policy  $\pi_{r_0, k_0}^*$ . Here  $V^*$  denotes the policy value,  $v^*$  the continuation value, and  $\Phi^*$  the log-sum-exp state value, where

$$\Phi_{r_0, k_0}^*(s) := \tau^* \log \left( \sum_{\tilde{a} \in \mathcal{A}^*} \exp \left( \tau^{*-1} \{ r_0(\tilde{a}, s) + \gamma v_{r_0, k_0}^*(\tilde{a}, s) \} \right) \right).$$

Let  $\rho_{r_0, k_0}^*$  and  $d_{r_0, k_0}^*$  denote the unnormalized discounted occupancy ratios in (12), with  $\pi = \pi_{r_0, k_0}^*$ . Define

$$z_{r_0, k_0}^*(a, s) := d_{r_0, k_0}^*(a, s) \{ q_{r_0, k_0}^*(a, s) - V_{r_0, k_0}^*(s) \}.$$

The total advantage-weighted state-action representer is

$$\tilde{d}_{r_0, k_0}^* := \{ (\mathcal{T}_{r_0, k_0}^*)^{-1} \}^* z_{r_0, k_0}^*, \quad \tilde{\rho}_{r_0, k_0}^*(s) := \sum_{a \in \mathcal{A}} \pi_0(a | s) \tilde{d}_{r_0, k_0}^*(a, s), \quad (13)$$

where the adjoint is taken in  $L^2\{\rho_0(s)\pi_0(a | s)\}$ . Equivalently, for every  $h$  in this space,

$$E_0\{\tilde{d}_{r_0, k_0}^*(A, S)h(A, S)\} = E_0[z_{r_0, k_0}^*(A, S)\{(\mathcal{T}_{r_0, k_0}^*)^{-1}h\}(A, S)].$$

This definition includes the current contribution  $z_{r_0, k_0}^*$ ; the successor-state occupancy construction used for estimation represents only its strictly future component. In the following condition, let  $p_{k_0, r_0}^* := p_{k_0}^{\pi_{r_0, k_0}^*, \gamma}$  denote the stationary  $\mu$ -density under  $\pi_{r_0, k_0}^*$ .

**(D2)** (*Stationary overlap for  $\pi_{r_0, k_0}^*$* ) There exist  $b, B \in (0, \infty)$  such that  $b \leq \frac{p_{k_0, r_0}^*(s)}{\rho_0(s)}$ .

$$\frac{\pi_{r_0, k_0}^*(a|s)}{\pi_0(a|s)} \leq B \text{ for } \lambda\text{-a.e. } (a, s).$$

**(D3)** (*Lipschitz continuity of the soft Bellman fixed point*) There exists  $L \in (0, \infty)$  such

that, for all  $k \in \mathcal{K}$ ,  $\|v_{r_0,k}^* - v_{r_0,k_0}^*\|_{L^2(\lambda)} \leq L\|k - k_0\|_{\mathcal{K}}$ .

Condition [D3](#) ensures pathwise differentiability of  $\Psi$  with respect to the kernel by controlling von Mises remainder terms of the form  $\|v_{r_0,k}^* - v_{r_0,k_0}^*\|_{L^2(\lambda)}$  in terms of  $\|k - k_0\|_{\mathcal{K}}$ .

**Theorem 10** (EIF for softmax policy). *Let  $m(a, s, r, k) := V_{r,k}^*(s)$ . Under [D2](#) and [D3](#), the parameter  $\Psi : P \mapsto E_P[V_{r_P,k_P}^*(S)]$  is pathwise differentiable at  $P_0$  with efficient influence function*

$$\begin{aligned} \chi_0(s, a, s') &= d_{r_0,k_0}^*(a, s) + \frac{1}{\tau^*} \tilde{d}_{r_0,k_0}^*(a, s) - \rho_{r_0,k_0}^*(s) - \frac{1}{\tau^*} \tilde{\rho}_{r_0,k_0}^*(s) \\ &\quad + \frac{\gamma}{\tau^*} \tilde{d}_{r_0,k_0}^*(a, s) \{ \Phi_{r_0,k_0}^*(s') - v_{r_0,k_0}^*(a, s) \} \\ &\quad + d_{r_0,k_0}^*(a, s) \left\{ r_0(a, s) + \gamma V_{r_0,k_0}^*(s') - q_{r_0,k_0}^*(a, s) \right\} \\ &\quad + V_{r_0,k_0}^*(s) - \Psi(P_0) \end{aligned}$$

Relative to Theorem [9](#), this EIF adds the total advantage-weighted representer and the Bellman term  $\frac{\gamma}{\tau^*} \tilde{d}_{r_0,k_0}^*(a, s) \{ \Phi_{r_0,k_0}^*(s') - v_{r_0,k_0}^*(a, s) \}$ . It also centers  $\tilde{d}_{r_0,k_0}^*$  under the behaviour policy. Both additions arise because the target policy depends on the soft Bellman solution  $v_{r_0,k_0}^*$ .

We now turn to Example [2](#). Write  $d_{0,k_0}^{\pi,\gamma'}(a, s) := \rho_0^{\pi,\gamma'}(s) \frac{\pi(a|s)}{\pi_0(a|s)}$  for the discounted state-action occupancy ratio of  $\pi$  at discount  $\gamma'$ , where  $\rho_0^{\pi,\gamma'}$  is the associated state occupancy ratio. Define  $r_{0,\nu,\gamma,g}(a, s) := g(s) + q_{r_0-g,k_0}^{\nu,\gamma}(a, s) - V_{r_0-g,k_0}^{\nu,\gamma}(s)$ , and abbreviate  $q_{r_0,k_0,\nu}^{\pi,\gamma'} := q_{r_0,\nu,\gamma,g,k_0}^{\pi,\gamma'}$  and  $V_{r_0,k_0,\nu}^{\pi,\gamma'}(s) := \sum_{a \in \mathcal{A}} \pi(a | s) q_{r_0,k_0,\nu}^{\pi,\gamma'}(a, s)$ . In addition to the stationary overlap condition from Condition [D1](#), applied here with target discount  $\gamma'$ , we require stationary overlap for the normalization policy  $\nu$ :

**(D4)** (*Stationary overlap for  $\nu$* ) There exist constants  $b, B \in (0, \infty)$  such that  $b \leq \frac{p_{k_0}^{\nu,\gamma}(s)}{\rho_0(s)}$ .

$$\frac{\nu(a|s)}{\pi_0(a|s)} \leq B \text{ for } \lambda\text{-a.e. } (a, s).$$

**Theorem 11** (EIF for policy value under reward normalization). *Let  $m(a, s, r, k) := \sum_{\tilde{a} \in \mathcal{A}} \pi(\tilde{a} \mid s) q_{r,k,\nu}^{\pi,\gamma'}(\tilde{a}, s)$ , where  $q_{r,k,\nu}^{\pi,\gamma'}$  is the  $Q$ -function of  $\pi$  at discount  $\gamma'$  under the  $\nu$ -normalized reward  $r_{r,k,\nu,\gamma,g}$ . Under [D1](#), [D4](#), and [C1\\*](#)-[C3\\*](#), the parameter  $\Psi : P \mapsto E_P[m(A, S, r_P, k_P)]$  is pathwise differentiable at  $P_0$  with efficient influence function*

$$\begin{aligned} \chi_0(s, a, s') &= \rho_0^{\pi,\gamma'}(s) \left( \frac{\pi(a \mid s)}{\pi_0(a \mid s)} - \frac{\nu(a \mid s)}{\pi_0(a \mid s)} \right) \\ &\quad + d_{0,k_0}^{\pi,\gamma'}(a, s) \left( 1 - \frac{\nu(a \mid s)}{\pi(a \mid s)} \right) \{ r_0(a, s) - g(s) + \gamma V_{r_0-g,k_0}^{\nu,\gamma}(s') - q_{r_0-g,k_0}^{\nu,\gamma}(a, s) \} \\ &\quad + d_{0,k_0}^{\pi,\gamma'}(a, s) \{ r_{0,\nu,\gamma,g}(a, s) + \gamma' V_{r_0,k_0,\nu}^{\pi,\gamma'}(s') - q_{r_0,k_0,\nu}^{\pi,\gamma'}(a, s) \} \\ &\quad + V_{r_0,k_0,\nu}^{\pi,\gamma'}(s) - \Psi(P_0). \end{aligned}$$

Theorem 11 retains the same two-part structure as Theorem 9, but adds a second Bellman term for the normalization map. The first augmentation accounts for estimating the normalized reward  $r_{0,\nu,\gamma,g}$  through the auxiliary value  $q_{r_0-g,k_0}^{\nu,\gamma}$ ; the second is the usual off-policy correction for evaluating  $\pi$  under that normalized reward.

Results for the additional examples from Section 4.2, namely softmax values under normalization and structural coefficients, are collected in Appendix A.

## 6.2 Estimators and asymptotic theory

We next give example-specific DML estimators that avoid direct estimation of the transition kernel  $k_0$ . Instead, they target higher-level nuisance functions that yield doubly robust von Mises expansions and simpler rate conditions under which Theorem 8 applies. We focus on Examples 1a, 1b and 2; additional estimators, conditions, and asymptotic results are deferred to Appendices B and B.

We revisit Example 1a. Let  $\pi_n := \exp\{r_n\}$ , let  $q_n^{\pi,\gamma}$  estimate  $q_{r_0,k_0}^{\pi,\gamma}$ , and define  $V_n^{\pi,\gamma}(s) := \sum_{\tilde{a} \in \mathcal{A}} \pi(\tilde{a} \mid s) q_n^{\pi,\gamma}(\tilde{a}, s)$ . Let  $\rho_n^{\pi,\gamma}$  estimate the state occupancy ratio  $\rho_0^{\pi,\gamma}$ . Our DML esti-

mator of  $\psi_0 = E_0[V_{r_0, k_0}^{\pi, \gamma}(S)]$  is

$$\begin{aligned} \psi_n := & \frac{1}{n} \sum_{i=1}^n V_n^{\pi, \gamma}(S_i) + \frac{1}{n} \sum_{i=1}^n \rho_n^{\pi, \gamma}(S_i) \frac{\pi(A_i | S_i)}{\pi_n(A_i | S_i)} \{r_n(A_i, S_i) + \gamma V_n^{\pi, \gamma}(S'_i) - q_n^{\pi, \gamma}(A_i, S_i)\} \\ & + \frac{1}{n} \sum_{i=1}^n \rho_n^{\pi, \gamma}(S_i) \left\{ \frac{\pi(A_i | S_i)}{\pi_n(A_i | S_i)} - 1 \right\}. \end{aligned}$$

The nuisance  $q_{r_0, k_0}^{\pi, \gamma}$  can be estimated by fitted  $Q$ -evaluation, i.e., iterative regression on Bellman targets (Ernst et al., 2005; Munos, 2005; Riedmiller, 2005; Munos and Szepesvári, 2008; Mnih et al., 2013; van der Laan and Kallus, 2025a). The ratio  $\rho_0^{\pi, \gamma}$  can be estimated by DICE-style, moment-matching, or saddle-point methods (Liu et al., 2018; Uehara et al., 2020; Nachum et al., 2019; Zhang et al., 2020; Wang et al., 2023; van der Laan et al., 2025b). For example,  $\rho_0^{\pi, \gamma} = \mathcal{T}_{k_0, \pi}(\tilde{\alpha}_0)$  (van der Laan et al., 2025b), where

$$\tilde{\alpha}_0 := \operatorname{argmin}_{\alpha \in L^2(\mu)} E_0[\{\mathcal{T}_{k_0, \pi}(\alpha)(A_0, S_0)\}^2 - 2\alpha(S_0)].$$

Thus one may estimate  $\rho_0^{\pi, \gamma}$  by replacing  $\mathcal{T}_{k_0, \pi}$  with  $\mathcal{T}_{k_n, \pi}$ , or by using an equivalent min-max formulation (Liu et al., 2018; Uehara et al., 2020; Dikkala et al., 2020; Kallus and Uehara, 2020). The next theorem gives rate conditions under which  $\psi_n$  is asymptotically linear and efficient.

$$\textbf{(E1)} \text{ (Nuisance rates)} \left\{ \|\rho_n^{\pi, \gamma} - \rho_0^{\pi, \gamma}\|_{L^2(\rho_0)} + \|r_n - r_0\|_{\mathcal{H}, r_0} \right\} \|\mathcal{T}_{k_0, \pi, \gamma}(q_n^{\pi, \gamma} - q_0^{\pi, \gamma})\|_{\mathcal{H}, r_0} = o_p(n^{-1/2}).$$

**Theorem 12** (Asymptotic linearity for Example 1a). *Assume D1, C5(b), and E1–E3. Then Theorem 8 holds with influence function  $\chi_0$  given in Theorem 9.*

For Example 1b, let  $r_n$  estimate  $r_0$ , let  $v_n^*$  estimate the soft Bellman fixed point  $v_{r_0, k_0}^*$ , and define  $\pi_n^*(a | s) \propto 1\{a \in \mathcal{A}^*\} \exp\{\frac{1}{\tau^*}\{r_n(a, s) + \gamma v_n^*(a, s)\}\}$ . Let  $q_n^*$  estimate  $q_{r_0, k_0}^*$ , set  $V_n^*(s) := \sum_{\tilde{a} \in \mathcal{A}^*} \pi_n^*(\tilde{a} | s) q_n^*(\tilde{a}, s)$ , and let  $d_n^*$  and  $\tilde{d}_n^*$  estimate the ordinary and total

advantage-weighted state–action representers. Define their state averages under the fitted behaviour policy by

$$\rho_n^\star(s) := \sum_{a \in \mathcal{A}} \pi_n(a \mid s) d_n^\star(a, s), \quad \tilde{\rho}_n^\star(s) := \sum_{a \in \mathcal{A}} \pi_n(a \mid s) \tilde{d}_n^\star(a, s).$$

The resulting estimator of  $\psi_0 = E_0\{V_{r_0, k_0}^\star(S)\}$  is

$$\begin{aligned} \psi_n := & \frac{1}{n} \sum_{i=1}^n V_n^\star(S_i) + \frac{1}{n} \sum_{i=1}^n d_n^\star(A_i, S_i) \{r_n(A_i, S_i) + \gamma V_n^\star(S'_i) - q_n^\star(A_i, S_i)\} \\ & + \frac{\gamma}{n\tau^\star} \sum_{i=1}^n \tilde{d}_n^\star(A_i, S_i) \left[ \tau^\star \log \left\{ \sum_{a \in \mathcal{A}^\star} \exp\left(\frac{1}{\tau^\star} \{r_n(a, S'_i) + \gamma v_n^\star(a, S'_i)\}\right) \right\} - v_n^\star(A_i, S_i) \right] \\ & + \frac{1}{n} \sum_{i=1}^n \left\{ d_n^\star(A_i, S_i) + \frac{1}{\tau^\star} \tilde{d}_n^\star(A_i, S_i) - \rho_n^\star(S_i) - \frac{1}{\tau^\star} \tilde{\rho}_n^\star(S_i) \right\}. \end{aligned}$$

The fixed point  $v_n^\star$  can be estimated by soft  $Q$ -iteration ([Haarnoja et al., 2017](#); [van der Laan and Kallus, 2025b](#)), and  $q_n^\star$  by fitted  $Q$ -evaluation under  $\pi_n^\star$ . For the signed representer, let  $z_n = d_n^\star(q_n^\star - V_n^\star)$ . A Jordan decomposition of  $z_n$  permits separate positive FORE fits initialized at successor states. Those fits estimate the strictly future term, which must be added to  $z_n$  to obtain the total  $\tilde{d}_n^\star$  in the estimator above.

**(E1\*)** (*Nuisance rates*) The following rates hold:

- (a)  $\|v_n^\star - v_{r_0, k_0}^\star\|_{\mathcal{H}, r_0} = o_p(n^{-1/4})$ .
- (b)  $\|\tilde{d}_n^\star - \tilde{d}_{r_0, k_0}^\star\|_{\mathcal{H}, r_0} (\|r_n - r_0\|_{\mathcal{H}, r_0} + \|v_n^\star - v_{r_0, k_0}^\star\|_{\mathcal{H}, r_0}) = o_p(n^{-1/2})$ .
- (c)  $(\|d_n^\star - d_{r_0, k_0}^\star\|_{\mathcal{H}, r_0} + \|r_n - r_0\|_{\mathcal{H}, r_0} + \|v_n^\star - v_{r_0, k_0}^\star\|_{\mathcal{H}, r_0}) \{ \|\mathcal{T}_{r_0, k_0}^\star(q_n^\star - q_{r_0, k_0}^\star)\|_{\mathcal{H}, r_0} + \|r_n - r_0\|_{\mathcal{H}, r_0} + \|v_n^\star - v_{r_0, k_0}^\star\|_{\mathcal{H}, r_0} \} = o_p(n^{-1/2})$ .

**Theorem 13** (Asymptotic linearity for Example 1b). Assume [D2](#), [C5\(b\)](#), and [E1\\*](#)–[E3\\*](#).

Then Theorem 8 holds with influence function  $\chi_0$  given in Theorem 10.

For Example 2. Let  $\pi_n := \exp\{r_n\}$ , let  $q_n^{\nu,\gamma}$  estimate  $q_{r_0-g,k_0}^{\nu,\gamma}$ , and define  $V_n^{\nu,\gamma}(s) := \sum_{\tilde{a} \in \mathcal{A}} \nu(\tilde{a} \mid s) q_n^{\nu,\gamma}(\tilde{a}, s)$  and  $r_{n,\nu,\gamma,g}(a, s) := g(s) + q_n^{\nu,\gamma}(a, s) - V_n^{\nu,\gamma}(s)$ . Let  $q_{n,\nu}^{\pi,\gamma'}$  estimate  $q_{r_0,k_0,\nu}^{\pi,\gamma'}$ , set  $V_{n,\nu}^{\pi,\gamma'}(s) := \sum_{\tilde{a} \in \mathcal{A}} \pi(\tilde{a} \mid s) q_{n,\nu}^{\pi,\gamma'}(\tilde{a}, s)$ , and let  $\rho_n^{\pi,\gamma'}$  estimate  $\rho_0^{\pi,\gamma'}$ . The corresponding estimator of  $\psi_0 = E_0[V_{r_0,k_0,\nu}^{\pi,\gamma'}(S)]$  is

$$\begin{aligned} \psi_n &:= \frac{1}{n} \sum_{i=1}^n V_{n,\nu}^{\pi,\gamma'}(S_i) \\ &+ \frac{1}{n} \sum_{i=1}^n \rho_n^{\pi,\gamma'}(S_i) \frac{\pi(A_i \mid S_i)}{\pi_n(A_i \mid S_i)} \{r_{n,\nu,\gamma,g}(A_i, S_i) + \gamma' V_{n,\nu}^{\pi,\gamma'}(S'_i) - q_{n,\nu}^{\pi,\gamma'}(A_i, S_i)\} \\ &+ \frac{1}{n} \sum_{i=1}^n \rho_n^{\pi,\gamma'}(S_i) \left( \frac{\pi(A_i \mid S_i)}{\pi_n(A_i \mid S_i)} - \frac{\nu(A_i \mid S_i)}{\pi_n(A_i \mid S_i)} \right) \{r_n(A_i, S_i) - g(S_i) + \gamma V_n^{\nu,\gamma}(S'_i) - q_n^{\nu,\gamma}(A_i, S_i)\} \\ &+ \frac{1}{n} \sum_{i=1}^n \rho_n^{\pi,\gamma'}(S_i) \left( \frac{\pi(A_i \mid S_i)}{\pi_n(A_i \mid S_i)} - \frac{\nu(A_i \mid S_i)}{\pi_n(A_i \mid S_i)} \right). \end{aligned}$$

**(E1\*\*)** (*Nuisance rates*) The following rate conditions hold:

- (a)  $\{\|\rho_n^{\pi,\gamma'} - \rho_0^{\pi,\gamma'}\|_{L^2(\rho_0)} + \|r_n - r_0\|_{\mathcal{H},r_0}\} \|\mathcal{T}_{k_0,\pi,\gamma'}(q_{n,\nu}^{\pi,\gamma'} - q_{0,\nu}^{\pi,\gamma'})\|_{\mathcal{H},r_0} = o_p(n^{-1/2})$ .
- (b)  $\|r_n - r_0\|_{\mathcal{H},r_0} \left\{ \|\mathcal{T}_{k_0,\nu,\gamma}(q_n^{\nu,\gamma} - q_0^{\nu,\gamma})\|_{\mathcal{H},r_0} + \|\mathcal{T}_{k_0,\pi,\gamma'}(q_{n,\nu}^{\pi,\gamma'} - q_{0,\nu}^{\pi,\gamma'})\|_{\mathcal{H},r_0} \right\} = o_p(n^{-1/2})$ .

**Theorem 14** (Asymptotic linearity for Example 2). *Assume D1, D4, C5(b), and E1\*\*–E3\*\*. Then Theorem 8 holds with influence function  $\chi_0$  given in Theorem 11.*

## 7 Simulation study

We study the estimators in a continuous-state treatment MDP with two state variables, four actions, nonlinear rewards, and Gaussian state transitions. Samples consist of independent stationary one-step transitions. The primary logging policy is a smooth state-dependent quadratic multinomial logit with controlled overlap; its probabilities are estimated rather than supplied to the estimator. We use five-fold cross-fitting and 300 predeclared Monte Carlo replications per design point. Example 1a uses  $n \in \{2500, 5000, 10000\}$ , whereas the

policy-adaptive Example 1b uses  $n \in \{25000, 50000, 100000\}$ . A labeled appendix stress test restores the original soft-optimal logging policy and its heavier ratio tails. In the first study, a fine grid supplies simulation truth and a common high-accuracy Bellman-evaluation nuisance, thereby isolating estimated policy and ratio error. Grid truth is never available to FORE or A-PBV. The data-fusion study instead estimates a smooth Gaussian transition model from the observed transition source and solves fitted-grid Bellman equations.

**Estimated-ratio inference.** For Examples 1a and 1b, discounted occupancy ratios are learned from the same transition sample by neural fitted occupancy-ratio evaluation (FORE). The implementation uses two-hidden-layer networks, minibatches of 512 transitions, five stochastic-gradient updates per outer iteration, and gradient and log-ratio caps of 10. To keep the largest design practical, neural ratio optimization uses a deterministic maximum of 20,000 outer-training rows per fold; logging-policy fitting and held-out inference continue to use all assigned rows. Hyperparameters are frozen in a separate pilot, and the number of fitted-ratio iterations in  $\{30, 100, 300\}$  is selected within each training fold by held-out adversarial pairwise Bellman validation (A-PBV). This criterion compares pairwise log ratios through observable occupancy-balance moments and therefore uses neither the oracle ratio nor the target estimand. For Example 1b, the advantage-weighted ratio is signed. We fit its positive and negative source measures separately by FORE and subtract the fitted components. The ordinary ratio and the two signed components are then refit on the full training fold before evaluating the untouched fold.

Table 1 compares the plug-in and D-IRL estimators by bias, Monte Carlo standard deviation, and RMSE across the 300 locked replications. For D-IRL we additionally report the mean estimated standard error, 95% Wald coverage, and mean interval width. Oracle-ratio evaluations are retained only as labeled appendix diagnostics.

All 1,800 estimated-ratio replications completed without a nonfinite ratio fit or any



Table 1: Estimated-ratio inference with neural FORE.

| Design | $n$    | Estimator | Bias   | MC SD | RMSE  | Mean SE | Coverage | Width | Fail. |
|--------|--------|-----------|--------|-------|-------|---------|----------|-------|-------|
| 1a     | 2500   | Plug-in   | -0.050 | 0.117 | 0.127 | —       | —        | —     | 0.000 |
|        |        | D-IRL     | 0.007  | 0.138 | 0.138 | 0.127   | 0.933    | 0.498 | 0.000 |
| 1a     | 5000   | Plug-in   | -0.017 | 0.084 | 0.086 | —       | —        | —     | 0.000 |
|        |        | D-IRL     | 0.010  | 0.091 | 0.091 | 0.087   | 0.937    | 0.341 | 0.000 |
| 1a     | 10000  | Plug-in   | -0.010 | 0.050 | 0.051 | —       | —        | —     | 0.000 |
|        |        | D-IRL     | 0.005  | 0.058 | 0.058 | 0.061   | 0.963    | 0.238 | 0.000 |
| 1b     | 25000  | Plug-in   | 0.006  | 0.018 | 0.019 | —       | —        | —     | 0.000 |
|        |        | D-IRL     | -0.005 | 0.019 | 0.020 | 0.019   | 0.927    | 0.074 | 0.000 |
| 1b     | 50000  | Plug-in   | 0.003  | 0.013 | 0.014 | —       | —        | —     | 0.000 |
|        |        | D-IRL     | -0.003 | 0.013 | 0.014 | 0.013   | 0.953    | 0.052 | 0.000 |
| 1b     | 100000 | Plug-in   | 0.002  | 0.009 | 0.009 | —       | —        | —     | 0.000 |
|        |        | D-IRL     | -0.001 | 0.009 | 0.009 | 0.009   | 0.933    | 0.037 | 0.000 |

log-ratio clipping. In Example 1a, D-IRL coverage is 93.3–96.3%; in the more variable policy-adaptive Example 1b, it is 92.7–95.3%. Every cell’s binomial 95% interval contains the nominal 95% level. Mean influence-function standard errors track Monte Carlo standard deviations to within 9% in Example 1a and 3% in Example 1b. At  $n = 2500$  in Example 1a, debiasing reduces the plug-in bias from  $-0.050$  to  $0.007$ , at the expected cost of additional correction variance. Across Example 1b, absolute D-IRL bias is at most  $0.0054$ . The fitted ratios have mean normalized mass  $1.000$ – $1.001$ , effective-sample fractions  $0.65$ – $0.93$ , and mean 99th percentiles  $12.8$ – $30.7$ , with no clipping or numerical failures.

**A practical data-fusion normalization.** The second study adapts Example 2 to a transition log and an abundant outcome-summary source. The transition log contains  $(S, A, S')$  and the known logging propensities, but no rewards. The auxiliary source contains only  $(S, Y)$  over the state domain, with action deliberately unavailable and

$$g(s) = E(Y \mid S = s) = \sum_a \nu(a \mid s) r_0(s, a),$$

where  $\nu$  is a known uniform randomized normalization policy. We generate one auxiliary sample of size  $10^6$  with broad covariate coverage, fit a small smooth tanh network selected

Table 2: Data-fusion estimation conditional on the frozen outcome source.

| $n$   | Estimator | Bias   | MC SD | RMSE  | Mean SE | Coverage | Width | Fail. |
|-------|-----------|--------|-------|-------|---------|----------|-------|-------|
| 2500  | Plug-in   | 0.008  | 0.015 | 0.017 | —       | —        | —     | 0.000 |
|       | D-IRL     | 0.003  | 0.075 | 0.075 | 0.073   | 0.947    | 0.285 | 0.000 |
| 5000  | Plug-in   | 0.006  | 0.009 | 0.011 | —       | —        | —     | 0.000 |
|       | D-IRL     | 0.002  | 0.050 | 0.050 | 0.051   | 0.960    | 0.200 | 0.000 |
| 10000 | Plug-in   | 0.005  | 0.006 | 0.008 | —       | —        | —     | 0.000 |
|       | D-IRL     | -0.001 | 0.039 | 0.039 | 0.036   | 0.937    | 0.140 | 0.000 |

by held-out outcome loss, ensemble three source-only refits, and freeze the resulting  $\hat{g}$  across transition-sample replications. The target discount is 0.8. Within each of five outer folds, a polynomial-sieve Gaussian transition model is tuned by held-out next-state loss, and neural FORE estimates the target occupancy ratio. We report point error and D-IRL interval coverage at  $n \in \{2500, 5000, 10000\}$ . The confidence intervals condition on the abundant frozen outcome source; joint uncertainty from estimating  $g$  is intentionally negligible by design and is not claimed to be covered by the present EIF.

All 900 data-fusion replications completed successfully. D-IRL bias ranges from  $-0.0008$  to  $0.0029$ , and coverage is 93.7–96.0%; again, each cell’s binomial 95% interval contains 95%. The mean standard error agrees closely with the Monte Carlo standard deviation at  $n = 2500$  and  $5000$  and is 8% smaller at  $n = 10000$ , matching that cell’s modest undercoverage. FORE retains unit normalized mass, an effective-sample fraction near 0.67, mean 99th percentiles between 13.2 and 13.6, and zero clipping or failures. The plug-in estimator has lower RMSE here because the deliberately abundant outcome source makes the frozen normalization regression exceptionally accurate; D-IRL adds the transition-sample correction needed for the reported confidence intervals.

## 8 Conclusion

We developed a semiparametric framework for inference in softmax IRL and dynamic discrete choice models with nonparametric rewards. By expressing a broad class of reward-dependent estimands as smooth functionals of the behaviour policy and transition kernel, deriving their efficient influence functions, and constructing automatic debiased ML estimators, the framework delivers valid and computationally tractable inference under flexible reward representations. It therefore connects classical inference for Gumbel-shock DDC models with modern machine-learning-based IRL, and provides a reusable template for new reward-dependent targets within the shared softmax structure. Natural extensions include other shock structures, broader normalization constraints, and nonhomogeneous or finite-horizon settings.

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